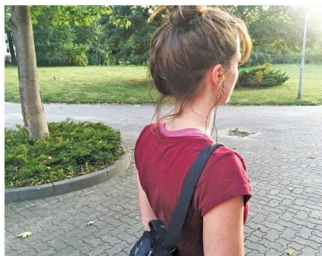


Mobile control with facial gestures

In everyday life, using mobile phones can be a challenge at times, especially when you are wearing gloves or have your hands full with purchases. Scientists from the Fraunhofer Institute for Computer Graphics Research (IGD) have come up with a solution: operation via head and face gestures, such as winking, smiling or nodding.



Test person with EarFS prototype controlling the mobile phone using facial gestures (Image courtesy: Fraunhofer IGD)

The EarFieldSensing (EarFS) device developed by the Fraunhofer IGD recognises the user's facial gestures via a special ear plug to allow interaction with the mobile phone. The ear plug sensor detects even the smallest movements in the face through the way the shape of the ear canal changes and measures muscle currents that arise during the movement of the face or head.

"The challenge was that these currents and movements are sometimes very small and need to be intensified," explained Denys Matthies, scientist at the Fraunhofer IGD. "In addition, the sensors cannot be interfered with by other movements of the body, such as vibrations during walking or external interferences. To solve this problem, an additional reference electrode was applied to the earlobe which records the signals coming from outside."

The signals detected inside the ear are matched with the signals coming from the outside—the remaining useful signal enables clear facial gesture identification, even if the person wearing the EarFS is in motion.

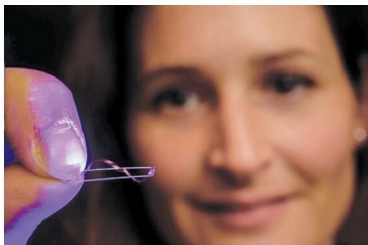
EarFS not only allows micro-interactions with a smartphone, such as accepting and rejecting phone calls or operating the music player, but also conclusions about weariness, exertion and other emotional states of the user. As a result, smartphones could warn drivers when signs of weariness and fatigue are registered or could switch automatically to mute when drivers are concentrating attentively.

New polymer goes for a walk when illuminated

Scientists at Eindhoven University of Technology and Kent State University have developed a new material that can undulate and therefore propel itself forward under the influence of light. To this end, they clamped a strip of this polymer material in a rectangular frame. When illuminated, it went for a walk all on its own.

This small device, the size of a paperclip, is the world's first machine to convert light directly into walking, simply using one fixed light source. The maximum speed is equivalent to that of a caterpillar, about half a centimetre per second. The researchers think it can be used to transport small items in hard-to-reach places or to keep the surface of solar cells clean. They placed grains of sand on the strip, which were removed by the undulating movement. The mechanism is so powerful that the strip can even transport an object that is much bigger and heavier than the device itself, uphill.

The motion of the new material is due to the fact that one side contracts in reaction to light, while the other one expands, causing it to bulge when illuminated. That deformation disappears instantaneously once the light is



TU Eindhoven researcher Anne Hélène Gélèbart showing the walking device (Image courtesy: Eindhoven University of Technology)

gone. Although the material looks transparent to the human eye, it fully absorbs the violet light the researchers used, thus creating a shadow behind it.

The research team managed to reach this specific behaviour of the material using 'liquid crystals.' The principle relies on the incorporation of a fast responding light-sensitive variant in a liquid crystalline polymer network. The researchers engineered a material such that this response is translated to an instantaneous deformation of the strip when illuminated, and relaxation directly when the light is gone.